Hardware Available at the Hackathon

We'll be stocked with computing platforms, sensors, radios, robotics, and electronics equipment to support rapid prototyping of search-and-rescue solutions.

Baseline Kit (Every Team)

Each team will have access to:

- **Raspberry Pi 5 (16 GB)** with 26 TOPS AI accelerator, storage, and power supply
- **Arduino-compatible microcontroller** with Grove expansion shield
- **Grove 40-in-1 sensor and actuator kit** with a variety of plug-and-play modules
- **Development accessories** for connecting sensors, prototyping circuits, and interfacing with hardware

Shared Hardware Library

Additional hardware will be available for teams to check out and share, including:

Category	Hardware
Additional Compute	Raspberry Pi Zero, ESP32, Arduino, NVIDIA Jetson, and other embedded development boards
Cameras & Vision	Standard, wide-angle, low-light/NoIR, AI-enabled, and depth cameras
Thermal & Infrared	Thermal imaging cameras and infrared sensors
LiDAR & Distance Sensing	2D/3D LiDAR, time-of-flight sensors, ultrasonic rangefinders
GPS & Localization	GPS/GNSS receivers, positioning modules, inertial sensors (IMUs), accelerometers, gyroscopes, magnetometers
RF & Wireless Communications	Software-defined radios (SDRs), Wi-Fi adapters, Bluetooth, LoRa, UWB ranging, and other wireless modules
Radar & Motion Detection	mmWave radar, Doppler radar, PIR motion sensors
Environmental Sensing	Temperature, humidity, pressure, gas, air-quality, light, and other environmental sensors

Audio & Acoustic Sensing	Microphones, speakers, acoustic sensors, and audio processing modules
Ground Robotics	Wheeled, tracked, and legged robots, robotic arms, and mobile robotics development kits
Aerial Robotics	Programmable microdrones, FPV drones, and drone development platforms
Motors & Actuators	DC motors, servos, motor controllers, pan-tilt mechanisms, and other actuators
Electronics & Prototyping	Breadboards, jumper wires, connectors, resistors, LEDs, switches, and other components
Test & Debug Equipment	Multimeters, oscilloscopes, logic analyzers, USB serial adapters, and bench power supplies
Fabrication & Assembly	Soldering stations, solder, wire, heat-shrink tubing, crimping tools, hand tools, and assembly supplies
Displays & Accessories	Portable monitors, USB hubs, power supplies, batteries, cables, and adapters